Explainable AI Clinical Decision Support System

Clinical Prediction Report

Condition: Heart Disease

Generated On: 14-09-2026 02:09:10

Patient Information

Field	Value
Patient ID	PAT003
Patient Name	Final Test Patient
Doctor / Clinician	Dr. Smith
Hospital / Clinic	Fortis Hospital
Disease	Heart Disease

Clinical Parameters

Feature	Observed Value
Age	65
Gender	Male
Chest Pain Type	Typical Angina
Resting Blood Pressure	160
Serum Cholesterol	300
Fasting Blood Sugar	Yes
Resting ECG	Left Ventricular Hypertrophy
Maximum Heart Rate	100
Exercise Angina	Yes
ST Depression	3.5
ST Segment Slope	Downsloping
Major Vessels	3
Thalassemia	Other

Model Assessment

Field	Value
Condition	Heart Disease

Field	Value
Prediction	Heart Disease
Class	Heart Disease
Probability	63.00%
Risk Level	Moderate
Clinical Recommendation	There are signs associated with heart disease. Consult a cardiologist and undergo additional clinical evaluation.

SHAP Evidence

Feature contribution to the model decision.

Feature	Contribution
Chest Pain Type	-0.1468
Major Vessels	0.0931
Maximum Heart Rate	0.0813
St Depression	0.0716
Thalassemia	-0.0680

LIME Evidence

Local explanation for this patient.

Feature / Rule	Weight
cp <= -0.16	-0.1562
thalach <= -0.62	0.0548
thal <= -0.88	-0.0473
oldpeak > -0.45	0.0466
-0.70 < exang <= 1.44	0.0341

SHAP-LIME Agreement

Consistency between the SHAP and LIME local explanation methods.

Field	Value
Agreement Score	86.38%
Feature Overlap	80.00%
Direction Agreement	100.00%
Ranking Agreement	81.25%

Field	Value
Overall Consistency	86.38%
Consistency Level	High
Shared Feature Count	4
Shared / Common Features	Chest Pain Type, Maximum Heart Rate, ST Depression, Thalassemia
SHAP Only	Major Vessels
LIME Only	Exercise Angina

Agreement measures consistency between SHAP and LIME explanations. It indicates explanation consistency, not clinical certainty.

Multi-Feature Counterfactual Explanation

Counterfactual explanations identify hypothetical alternative patient profiles in which one or more clinical features are changed to determine whether the model prediction can change.

Field	Value
Original Prediction	Heart Disease
Original Probability	63.00%

Counterfactual #1

Field	Value
New Prediction	No Heart Disease
New Probability	48.00%
Features Changed	2
Change Magnitude	0.73
Changed Clinical Features	Age: 65 → 52 Maximum Heart Rate: 100 → 153

Counterfactual #2

Field	Value
New Prediction	No Heart Disease
New Probability	48.00%
Features Changed	2
Change Magnitude	0.78
Changed Clinical Features	Age: 65 → 49 Maximum Heart Rate: 100 → 153

Counterfactual #3

Field	Value
New Prediction	No Heart Disease
New Probability	50.00%
Features Changed	2
Change Magnitude	0.80
Changed Clinical Features	Age: 65 → 52 Maximum Heart Rate: 100 → 160

Counterfactual #4

Field	Value
New Prediction	No Heart Disease
New Probability	50.00%
Features Changed	2
Change Magnitude	0.85
Changed Clinical Features	Age: 65 → 49 Maximum Heart Rate: 100 → 160

Counterfactual #5

Field	Value
New Prediction	No Heart Disease
New Probability	48.00%
Features Changed	2
Change Magnitude	0.93
Changed Clinical Features	ST Depression: 3.5 → 2.0925 Maximum Heart Rate: 100 → 153

What does this mean?

Counterfactual explanations show hypothetical patient profiles in which one or more model input features are changed and the resulting prediction is evaluated. These results illustrate the model's decision boundary.

Important: Counterfactual scenarios are model-generated hypothetical examples. They must not be interpreted as clinical treatment targets, prescriptions, diagnoses, or instructions to change a patient's health measurements.

Clinical Interpretation Note

The prediction and explanation values in this report represent the output of the machine-learning system for the submitted patient data. SHAP and LIME provide model interpretability, while counterfactual analysis illustrates hypothetical changes that may cross the model's decision boundary.

Disclaimer: This report is intended for clinical decision support only. Model outputs, SHAP values, LIME explanations, agreement scores, and counterfactual scenarios must not be interpreted as definitive diagnoses, prescriptions, treatment instructions, or required clinical targets. Final clinical decisions must be made by qualified healthcare professionals using appropriate clinical assessment and additional evidence.

Generated by Explainable AI Clinical Decision Support System

Doctor Signature

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Dr. Smith

Patient Signature

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Final Test Patient